

PhD in Civil Engineering – CV

Bagher Nikoufar

Tel: +98 (0)4133241587

Mob: +98 (0)9141167102

E-mail: baghernik@yahoo.com

Date of Birth: Aug 23rd, 1973

https://www.researchgate.net/profile/Bagher-Nikoufar?ev=hdr_xprf

www.linkedin.com/in/bagher-nikoufar-b325a343

Education

Sep 2021 – Jan 2025

University of Tabriz, Tabriz, Iran.

PhD of Civil Engineering, Water Resources Engineering, GPA: 18.85/20.

Thesis Title: Application of Z-Numbers for the Optimizing Operation of Reservoirs.

Sep 1999 – July 2002

Sahand University of Technology, Tabriz, Iran.

M.Sc. of Civil Engineering - Marine Structures, GPA: 15.95/20.

Thesis Title: Two-Dimensional Numerical Model of Sedimentation in Dam Reservoirs (Case Study of Bukan Dam).

Sep 1992 – Aug 1996

Power and Water University of Technology(Shahid Abbaspour), Tehran, Iran.

B.Sc. of Civil Engineering – Dam and Network Operation , GPA: 15.13/20.

Work Experience

Nov 2003 – Present

East Azerbaijan Regional Water Company, Tabriz, Iran.

Position: Head of Operation and Maintenance of Dams and Hydropower Plants.

Duties:

- Compilation of the management plan for resources and consumption of dams in the province.
- Supervision of restoration and reconstruction projects and repairs of dams.
- Behavioral measurement and behavior of dams.

- Coordination and supervision of water supply for beneficiaries in the dams.

Jan 2014 – Jan 2018

East Azerbaijan Regional Water Company, Tabriz, Iran.

Position: Project Manager of Feasibility and Finding the Potential of Building Small Hydropower Plants.

Duties:

- Study of water energy potential.
- Identification and potential location of small power plants.
- Researching the history of the use of water energy potential.
- Supervision of the implementation of small power plant.

Sep 2014 – Oct 2018

East Azerbaijan Regional Water Company, Tabriz, Iran.

Position: Head of Surface Water Operation Working Group.

Duties:

- Collecting new ideas in surface water to improve and develop the work.
- Collection of proposed research projects.
- Supervision of research projects.
- Operationalizing research and evaluating its effects.

Sep 2008 – March 2011

Ajabshir Islamic Azad University, Ajabshir, Iran.

Position: Lecturer.

Duties:

- Teaching Hydraulics.
- Teaching Fluid Mechanics.
- Strength of Materials.
- Concrete Structures.
- Teaching Static.

Sep 2003 – March 2008

Maragheh Islamic Azad University, Maragheh, Iran.

Position: Lecturer.

Duties:

- Teaching Hydraulics.
- Teaching Fluid Mechanics.
- Teaching Static.

Nov 1997 – Nov 2003

Zarine Roud Irrigation Network Operating Company, Miyandoab, Iran.

Position: Responsible for Planning and Maintenance of Irrigation Networks.

Duties:

- Irrigation network operation.
- Maintenance repairs and planning.
- Programming and organizing subscriber affairs services.

Research Publications

1. Shiri.J, Kisi.O, **Nikoofar.B**, 2012, 'Forecasting Daily Lake Levels Using Artificial Intelligence Approaches', *Computers & Geosciences for Publication in Journal of Elsevier*, Vol. 41, pp: 169-180. DOI: [10.1016/j.cageo.2011.08.027](https://doi.org/10.1016/j.cageo.2011.08.027)
2. Kisi.O, Shiri.J, Makarynsky .O, Shiri.A.A, **Nikoofar.B**, 2012, 'Forecasting daily stream flows using artificial intelligence approaches', *ISH Journal of Hydraulic Engineering in Journal of Elsevier*, Vol.18, pp: 204-214. DOI: [10.1080/09715010.2012.721189](https://doi.org/10.1080/09715010.2012.721189)
3. **Nikoufar.B** (Corresponding Author), Mahtabi.G, Faregh.A, 2013, 'Optimal Operation of Water Resources of Storage Reservoirs by Controlling Seasonal Floods, Alavian Dam Case Study', *Journal of Basic and Applied Scientific Research*, Article No. JBASR-622-2.
Link: <https://www.semanticscholar.org/paper/Optimal-Operation-of-Water-Resources-of-Storage-by-Nikoufar-Mahtabi/74f0d546a7dc239afe8c781269d3383de6785f78>
4. Najafi.H, **Nikoufar.B**, Nourani.V, Behfar.N, 2024, 'Forecasting Reservoir Inflow and Employing the Combined Genetics-Particle Swarming Optimization Approach for Alavian Reservoir Operation', *Journal of Civil and Environmental Engineering*, Tabriz, Iran. DOI: [10.22034/ceej.2024.60171.2320](https://doi.org/10.22034/ceej.2024.60171.2320)
5. **Nikoufar.B** (Corresponding Author), Nourani.V, Hoseini. B.A, 2024, 'Determining the optimal operation parameters of Alavian dam using the combination of genetic and particle swarm algorithms', *Journal of Water and Hydraulics*, Tabriz, Iran, 34(4), pp:89-110. DOI: [10.22034/hws.2024.18918](https://doi.org/10.22034/hws.2024.18918)
6. Nourani.V, **Nikoufar.B** (Corresponding Author), Ng.A, Jabbarian.P.N, Gökçekuş.H, 2024 'Uncertainty Assessment of Ensemble Base Machine Learning Modeling for Multi-step Ahead Forecasting of Dam Reservoir Inflows', *Journal of Springer*, 48(1). <https://doi.org/10.1007/s40996-024-01685-2>
7. **Nikoufar.B** (Corresponding Author), Nourani.V, 2024, 'Using Z-Numbers in Determine Optimal Operation Parameters of Dam Reservoirs', *Iranian Journal of Irrigation & Drainage*, ISSN:2008-7942. https://idj.iaid.ir/article_211793.html?lang=en
8. Nourani.V, Najafi.H, **Nikoufar.B**, Jinhui Huang,J, 2025, 'Optimizing Reservoir Operations with Z-Numbers: Addressing Uncertainty and Reliability', *Journal of Hydrology*.

9. Sadikoglu.F, **Nikoufar.B** , Nourani.V, 2025, ‘Optimizing Reservoir Operations with Fuzzy Logic: Addressing Uncertainty’, Journal of Springer, pp:199-208, https://link.springer.com/chapter/10.1007/978-3-031-98565-2_23
10. **Nikoufar.B** , Nourani.V, Hoseini.B.A, 2026, ‘Predictive Modeling of Reservoir Inflow for Multi-Step Ahead to Improve Management of Alavian Dam Reservoir’, Journal of Civil and Environmental Engineering University of Tabriz, Vol:56, ISSN: 2008-7918. DOI: [10.22034/CEEJ.2024.59627.2308](https://doi.org/10.22034/CEEJ.2024.59627.2308)
11. Najafi.H, Huang. JJ, Nourani.V, Behfar.N, **Nikoufar.B**, 2026, ‘An Interpretable AI-Based Decision-Support System for Adaptive Reservoir Operation under Climate Stress’, Journal of Hydrology. DOI:[10.2139/ssrn.6765997](https://doi.org/10.2139/ssrn.6765997)
12. **Nikoufar.B** (Corresponding Author) , Nourani.V, Kantoush.S.A, Jabbarian.N, 2026, ‘ Application of Z-Numbers in the Optimal Allocation of Environmental Water to Lake Urmia from Dam Reservoirs’, Journal of Hydrology.([Under review](#))

Conferences

1. **Nikoufar, B**, Najafi, H, Nourani, V, Shiva, E, Huang, J, May 2025, ‘Hybrid Models for Multi-Step Ahead Estimation of Optimal Parameters in Dam Reservoir Operations’, ICOLD-CIGB, Chengdu, China
2. **Nikoufar, B**, Jun 2023, ‘Multi-step Modeling of Inflow to Alavian Dam Reservoir with Improve Dam Reservoir Management’, 91st Annual ICOLD Meeting, Gothenburg, Sweden.
3. **Nikoufar, B**, Mar 2018, ‘The Use of Sea Wave Energy in the Study and Design of Micro-scale Linear Water Turbine’, Iranian Conference on Renewable Energy, Shahid Madani University of Tabriz, Tabriz, Iran.
4. **Nikoufar, B**, Jan 2013, ‘Application of Ribasim Model in Water Resources Management (Case Study of Sufi Chai River Basin)’, National Conference on Construction and Sustainable Development, Mashhad, Iran.
5. **Nikoufar, B**, Aug 2011, ‘Establishing a System of Preventive Maintenance of Dams (Case Study of Aras Dam)’, The 1st International Conference of Electric Power Plants, Tehran University, Tehran, Iran.
6. **Nikoufar, B**, Oct 2009, ‘Reuse of Treated Wastewater as a Non-conventional Water Source’, The 3rd Specialized Conference and Exhibition of Environmental Engineering, Tehran University, Tehran, Iran.

7. **Nikoufar, B**, Oct 2008, 'Establishment of Information Management System Using Information Systems in Reservoirs', *The 3rd Conference of Water Resources Management*, Tabriz University, Tabriz, Iran.
8. **Nikoufar, B**, Apr 2007, 'Revision of the Rule Curve for the Use from the Alavian Dam Reservoir for Flood Control and Improving the Supply of the Downstream Needs of the Dam', *The 3rd National Congress on Civil Engineering*, Tabriz, Iran.
9. **Nikoufar, B**, Sep 2007, 'Repair and Renovation of Concrete Structures Using Polyurethane Chemicals', *The 6th Iranian Hydraulic Conference*, Shahr Kurd University, Iran.
10. **Nikoufar, B**, Jan 2002, 'Investigating Sedimentation in Reservoirs of Dams Using Two-dimensional Numerical Analysis', *The 6th International Seminar on River Engineering*, Shahid Chamran University, Ahvaz, Iran.
11. Sadikoglu.F, **Nikoufar.B** (Corresponding Author), Nourani.V, Khaleghi,H, Jabbarian.P.N, 2025 'Modeling for Multi-Step Ahead Forecasting of Dam Reservoir Inflows for Improved Reservoir Management', 2nd International Conference on Smart Environment and green Technologies ICSEGT2025.
12. **Nikoufar.B**(Corresponding Author), Najafi.H, Nourani.V, Ebadi.S, Huang. JJ, 2025, 'Hybrid Models for Multi-Step Ahead Estimation of Optimal Parameters in Dam Reservoir Operations', International Symposium, ICOLD-CIGB 2025, Chengdu, China.

Research Activities

- **Aug 2013 - Aug 2016** : East Azarbaijan Water Org, Tabriz, Iran.
Title of Project: How to Evaluate and Estimate the Technical and Engineering Services of and Maintenance of Dams.
- **June 2022 – Sep2024:** Tabriz University with East Azarbaijan Water Org, Tabriz, Iran
Title of Project: Development and updating of operation rule curves of Alavian and Qaleh Chay dams (dams of Urmia Lake basin) as a supplier of different water needs using mathematical models
- **June 2023 – May 2025:** Tabriz University with East Azarbaijan Water Org, Tabriz, Iran
Title of Project: Increasing the resilience of supply systems, water supply and exploitation of water resources and facilities in East Azarbaijan province with a passive defense approach
- **June 2024 – Sep2026:** Tabriz University with East Azarbaijan Water Org, Tabriz, Iran
Title of Project: Feasibility Study of Deployment and Operation of Artificial Intelligence Tools in East Azerbaijan Regional Water Company
- **May 2026 – Present:** Tabriz University with East Azarbaijan Water Org, Tabriz, Iran
Title of Project: Optimization of Reservoir Operation Parameters for Downstream Water Demand Supply Using Artificial Intelligence and Fuzzy Logic (Case Study: Nahand and Sattarkhan Dams)

Description:

In the above five studies, I participated as a research associate with **Dr. Vahid Noorani** (Professor at Tabriz University) and **Dr. Nazila Kardan** (Professor at Madani University) and I can obtain a letter of confirmation of cooperation from the respected professor and from the employer (East Azerbaijan Regional Water Company).

Book Publication

May 2025 - July 2026

Title of Book: Application of Artificial Intelligence with Z-Number Theory in Water

Resource Management Optimization and Dam Reservoir Operation(Book, in press).

Patent

June 2012

Title of Patent: Design and Construction of a Linear Synchronous Generator that Produces Electricity from the Energy of Water Waves.

Place of Registration: General Directorate of Industrial Property, Iran Patents Department.

Certificates

- Dec 2021: **Remote Control in Water Facilities**, Scientific and Applied Higher Education Institute of Water and Electricity Industry, Tabriz, Iran.
- March 2021: **Work with WEAP Software**, Scientific and Applied Higher Education Institute of Water and Electricity Industry, Tabriz, Iran.
- Dec 2020: **Advanced Methods in Water Resources Management Planning**, Scientific and Applied Higher Education Institute of Water and Electricity Industry, Tabriz, Iran.
- April 2017, Workshop **The Water Reform Journey in Australia**, Organised jointly by The University of Melbourne and Sharif University of Technology in collaboration with 10th National Congress on Civil Engineering and UNESCO Chair at Sharif University, Tehran, Iran.
- June 2016: **Examining and Explaining the Issues of Seismic Control of Structures with a Focus on Intelligent Structures**, Education and Research Center for Development and Foresight of Management and Planning Organization of East Azarbaijan Province, Tabriz, Iran.
- Aug 2015: **Integrated Management of Power Plant Reservoirs with the Goal of Optimal Energy**, Scientific and Applied Higher Education Institute of Water and Electricity Industry, Tabriz, Iran.
- Aug 2015: **ArcGis and GIS Software Training**, Scientific and Applied Higher Education Institute of Water and Electricity Industry, Tabriz, Iran.

- Aug 2005 - Aug 2006: **Mike11 HD Modelling and Mike 11 Reservoir Modelling**, DHI Company, Tabriz, Iran.
- Aug 2005: **Integrated Water Resources Management**, International Institute for GEO-Netherlands, Tabriz, Iran.
- Aug 2003, **Advanced Visual Basic Programming**, Scientific and Applied Higher Education Institute of Water and Electricity Industry, Tabriz, Iran.

Research Interests

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| • Water Resources Modeling and Prediction | • Optimizing the Operation of Reservoirs of Dams |
| • Planning and Decision Making in Water Resources Management | • Fuzzy Logic and Z Number |
| • Deep Learning and Machine Learning | • Smart Water |
| • Remote Sensing and Teleconnections | • Hydroinformatics |

Professional Memberships

- **June 2021 – Present:** The main member of **the Committee of Hydrography and Sedimentation of Iran.**
- **Sep 2014 – Present:** The main member of **the Committee of the Surface Water Protection and Exploitation of East Azerbaijan Regional Water Company.**
- **Nov 2006 – Present:** The main member of **the National Committee of Great Dams of Iran.**

Software Skills

- **Programming:** Matlab, Fortran, GAMS, Python, Lingo.
- **Technical:** ArcMap, Surfer, AutoCAD, SPSS, VENSIM, WEKA, WEAP, HEC-HMS, HEC-RAS, Google Earth Engine, Mike21, GeoStudio.
- **General:** MS Office.

Language Skills

- **Persian:** Native;
- **Azerbaijani:** Native;
- **English:** Advanced.

Referees

- **Dr. Vahid Nourani**

- Prof. of Civil & Environmental Eng.,
Director of Center of Excellence in Hydroinformatics,
Faculty of Civil Eng.,
University of Tabriz, Iran.
- Adjunct Prof.,
Faculty of Civil & Environmental Eng.,
Near East University, Cyprus.
- Honorary Adjunct Prof.,
College of Eng., IT and Environment,
Charles Darwin University, Australia.

E-mail: nourani@tabrizu.ac.ir and vnourani@yahoo.com

Tel: +98 (0) 9144030332

- **Dr. Mohammad Ali Lotfollahi Yaghin**

Prof. of Civil & Environmental Eng.,
Faculty of Civil Eng.,
University of Tabriz,
Tabriz, Iran.

E-mail: lotfollahi@tabrizu.ac.ir

Tel: +98 (0) 09144186490

- **Dr. Elnaz Sharghi**

Associate Prof. of Civil & Environmental Eng.,
Faculty of Civil Eng.,
University of Tabriz,
Tabriz, Iran.

E-mail: sharghi@tabrizu.ac.ir and elnaz_sharghi@yahoo.com

Tel: +98 (0) 9144178419

- **Dr. MohammadTaghi Sattari**

Associate Prof.,
Department of Water Engineering,
Faculty of Agriculture,
University of Tabriz,
Tabriz, Iran.

E-mail: mtsattar@tabrizu.ac.ir and mtsattar@gmail.com

Tel: +98 (0) 9143044566